

Solar N appendix

Sun · N · downloadable appendix

N.json v1.14 · 12 visible products · 0 quarantined entries

products

Independent admitted products; no combined headline abundance.

Highlighted Products

No adopted highlighted product has been selected. See the independent products below.

Error-bar Forest Plot

red-optical — iag_fts_solar_atlas solar_iag A(N) dex; axis 7.83-8.50		
SET-AGSS21_ENGINE-A · N I · Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · line set not supplied Reference Grade · ALL · n=2		8.091 ±0.107 stat ±0.042 syst
SET-AGSS21 · N I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Reference Grade · ALL · n=4		8.351 ±0.119 stat ±0.042 syst
SET-AGSS21 · N I · Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · line set not supplied Reference Grade · ALL · n=2		8.091 ±0.107 stat ±0.042 syst
red-optical — kpno_solar_atlas solar_kpno_kurucz2005_corrected A(N) dex; axis 7.83-8.50		
SET-AGSS21_ENGINE-A · N I · Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · line set not supplied Reference Grade · ALL · n=2		8.189 ±0.027 stat ±0.042 syst
SET-AGSS21 · N I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Reference Grade · ALL · n=4		8.293 ±0.069 stat ±0.042 syst
SET-AGSS21 · N I · Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · line set not supplied Reference Grade · ALL · n=2		8.189 ±0.027 stat ±0.042 syst
red-optical — kpno_solar_atlas solar_kpno_molecfit_corrected A(N) dex; axis 7.83-8.50		
SET-AGSS21_ENGINE-A · N I · Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · line set not supplied Reference Grade · ALL · n=2		8.016 ±0.155 stat ±0.042 syst
SET-AGSS21 · N I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Reference Grade · ALL · n=4		8.287 ±0.122 stat ±0.042 syst
SET-AGSS21 · N I · Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · line set not supplied Reference Grade · ALL · n=2		8.016 ±0.155 stat ±0.042 syst

red-optical axis: 7.83 · 8.00 · 8.17 · 8.33 · 8.50 A(N) dex

VIS — harps_solar_harps_molecfits_corrected A(N) dex; axis 6.52-8.25

MOL-CN_red · N I · Synth · 1D-LTE · SYNTH · 1D-LTE ·
line set not supplied
Codex Grade · ALL · n=194



7.384
±0.835 stat ±0.768 syst

VIS axis: 6.52 · 6.95 · 7.38 · 7.82 · 8.25 A(N) dex

NIR — iag_fts_solar_atlas_solar_iag A(N) dex; axis 7.93-8.31

MOL-CN_AX_IR · N I · Synth · 1D-LTE · SYNTH ·
1D-LTE · line set not supplied
Codex Grade · ALL · n=244



7.999
±0.037 stat ±0.033 syst

NIR — kpno_solar_atlas_solar_kpno_molecfits_corrected A(N) dex; axis 7.93-8.31

MOL-CN_AX_IR · N I · Synth · 1D-LTE · SYNTH ·
1D-LTE · line set not supplied
Codex Grade · ALL · n=215



8.233
±0.046 stat ±0.036 syst

NIR axis: 7.93 · 8.03 · 8.12 · 8.21 · 8.31 A(N) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Solid bars show the supplied statistical uncertainty; wireframes show the supplied systematic uncertainty. The statistical basis and any incomplete error budget remain attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

Solar nitrogen context

The nitrogen strategy uses red-optical N I as the primary atomic route, with NH providing an independent molecular check and CN a carbon-dependent cross-check. An unresolved or unconstrained fit supplies no adopted Solar nitrogen abundance.

[C/O indicator strategy](#) · [Methodology](#) · [Nitrogen strategy](#)

Product provenance and caveats

SET-AGSS21_ENGINE-A · N I · Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · line set not supplied: {"display": "Synth · 1D-NLTE · Bergemann", "dominant_term": "line-to-line scatter", "gf_pool_graded": true, "gf_rung_summary": "gf rung 2 (gf scale (cited NIST class))", "grade": "Reference Grade", "n_excluded": 2, "n_lines": 2, "provenance": {"artifact_mtime": "2026-09-12T21:16:43Z", "copied_to": "data/results/band_products/NI_6910_9199_iag_fts_solar_atlas_solar_iag_SYNTH_SET-AGSS21_ENGINE-A_products.csv", "host": "Sirius", "ingested_at": "2026-09-12T23:16:46Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/NI_6910_9199_iag_fts_solar_atlas_solar_iag_SYNTH_SET-AGSS21_ENGINE-A_products.csv", "sha256": "aa82ee1637013b3b34a69f55c16d2af69180a9539b67406a972ddda1d9cda621"}, "sigma_stat": 0.107, "sigma_syst": 0.042, "stat_basis": "measured — RMS of the random terms, 0.10705 dex at n_lines=2"}

SET-AGSS21 · N I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": "line-to-line scatter", "gf_pool_graded": true, "gf_rung_summary": "gf rung 2 (gf scale (cited NIST class))", "grade": "Reference Grade", "n_excluded": 0, "n_lines": 4, "provenance": {"artifact_mtime": "2026-09-12T21:16:43Z", "copied_to": "data/results/band_products/NI_6910_9199_iag_fts_solar_atlas_solar_iag_SYNTH_SET-AGSS21_products.csv", "host": "Sirius", "ingested_at": "2026-09-12T23:16:47Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/NI_6910_9199_iag_fts_solar_atlas_solar_iag_SYNTH_SET-AGSS21_products.csv", "sha256": "d1fbf12d550732231df32e167ae4832ec0e079c4004546f93caff0540e3784c7"}, "sigma_stat": 0.119, "sigma_syst": 0.042, "stat_basis": "measured — RMS of the random terms, 0.11895 dex at n_lines=4"}

SET-AGSS21_ENGINE-A · N I · Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · line set not supplied: {"display": "Synth · 1D-NLTE · Bergemann", "dominant_term": "telluric residual", "gf_pool_graded": true, "gf_rung_summary": "gf rung 2 (gf scale (cited NIST class))", "grade": "Reference Grade", "n_excluded": 2, "n_lines": 2, "provenance": {"artifact_mtime": "2026-09-12T20:55:38Z", "copied_to": "data/results/band_products/NI_6910_9199_kpno_solar_atlas_solar_kpno_kurucz2005_corrected_SYNTH_SET-AGSS21_ENGINE-A_products.csv", "host": "Sirius", "ingested_at": "2026-09-12T23:16:47Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/NI_6910_9199_kpno_solar_atlas_solar_kpno_kurucz2005_corrected_SYNTH_SET-AGSS21_ENGINE-A_products.csv", "sha256": "ae17208b1ffcd06e5eaf9c8971adc16630c14485955e6cbdaaac4719a92a34ce"}, "sigma_stat": 0.0266, "sigma_syst": 0.042, "stat_basis": "measured — RMS of the random terms, 0.02655 dex at n_lines=2"}

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SET-AGSS21 · N I · Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · line set not supplied: {"display": "Synth · 1D-NLTE · Bergemann", "dominant_term": "line-to-line scatter", "grade": "Reference Grade", "n_excluded": 2, "n_lines": 2, "provenance": {"artifact_mtime": "2026-09-12T21:16:43Z", "copied_to": "data/results/band_products/NI_6910_9199_iag_fts_solar_atlas_solar_iag_SYNTH_SET-AGSS21_ENGINE-A_products.csv", "host": "Sirius", "ingested_at": "2026-09-13T04:26:39Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/NI_6910_9199_iag_fts_solar_atlas_solar_iag_SYNTH_SET-AGSS21_ENGINE-A_products.csv", "sha256": "aa82ee1637013b3b34a69f55c16d2af69180a9539b67406a972ddda1d9cda621"}, "sigma_stat": 0.107, "sigma_syst": 0.042, "stat_basis": "measured — RMS of the random terms, 0.10705 dex at n_lines=2"}

SET-AGSS21 · N I · Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · line set not supplied: {"display": "Synth · 1D-NLTE · Bergemann", "dominant_term": "telluric residual", "grade": "Reference Grade", "n_excluded": 2, "n_lines": 2, "provenance": {"artifact_mtime": "2026-09-12T20:55:38Z", "copied_to": "data/results/band_products/NI_6910_9199_kpno_solar_atlas_solar_kpno_kurucz2005_corrected_SYNTH_SET-AGSS21_ENGINE-A_products.csv", "host": "Sirius", "ingested_at": "2026-09-13T04:26:40Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/NI_6910_9199_kpno_solar_atlas_solar_kpno_kurucz2005_corrected_SYNTH_SET-AGSS21_ENGINE-A_products.csv", "sha256": "ae17208b1ffcd06e5eaf9c8971adc16630c14485955e6cbdaaac4719a92a34ce"}, "sigma_stat": 0.0266, "sigma_syst": 0.042, "stat_basis": "measured — RMS of the random terms, 0.02655 dex at n_lines=2"}

SET-AGSS21 · N I · Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · line set not supplied: {"display": "Synth · 1D-NLTE · Bergemann", "dominant_term": "line-to-line scatter", "grade": "Reference Grade", "n_excluded": 2, "n_lines": 2, "provenance": {"artifact_mtime": "2026-09-12T21:06:20Z", "copied_to": "data/results/band_products/NI_6910_9199_kpno_solar_atlas_solar_kpno_molecfits_corrected_SYNTH_SET-AGSS21_ENGINE-A_products.csv", "host": "Sirius", "ingested_at": "2026-09-13T04:26:40Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/NI_6910_9199_kpno_solar_atlas_solar_kpno_molecfits_corrected_SYNTH_SET-AGSS21_ENGINE-A_products.csv", "sha256": "a88325dceb9770b20cbbecff32bc8b8e4776116a1214f414906da4b367fa6e61"}, "sigma_stat": 0.1551, "sigma_syst": 0.042, "stat_basis": "measured — RMS of the random terms, 0.15505 dex at n_lines=2"}

MOL-CN_red · N I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": null, "grade": "Codex Grade", "n_excluded": 0, "n_lines": 194, "provenance": {"artifact_mtime": "2026-09-13T04:18:40Z", "copied_to": "data/results/band_products/NI_MOL-CN_red_harps_solar_harps_molecfits_corrected_SYNTH_MOL-CN_red_products.csv", "host": "Sirius", "ingested_at": "2026-09-13T04:21:45Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/NI_MOL-CN_red_harps_solar_harps_molecfits_corrected_SYNTH_MOL-CN_red_products.csv", "sha256": "3693da7ea8f6695a379152368be123200b8d9183b9f88cf8b364384348cdcbe5"}, "sigma_stat": 0.835, "sigma_syst": 0.768, "stat_basis": "measured — 1 sigma from the chi2 curvature of THIS band's fit (pipeline.fit_constraint.curvature_sigma, rescaled to red_chi2 = 1); NaN where the curvature was not measurable, never 0 and never the retired 1.000 clip"}

MOL-CN_AX_IR · N I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": null, "grade": "Codex Grade", "n_excluded": 0, "n_lines": 244, "provenance": {"artifact_mtime": "2026-09-13T16:56:48Z", "copied_to": "data/results/band_products/NI_MOL-CN_AX_IR_iags_fts_solar_atlas_solar_iag_SYNTH_MOL-CN_AX_IR_products.csv", "host": "Sirius", "ingested_at": "2026-09-13T16:57:05Z", "path": "/home/damienabraxas/rya1214_repo/data/results/band_products/NI_MOL-CN_AX_IR_iags_fts_solar_atlas_solar_iag_SYNTH_MOL-CN_AX_IR_products.csv", "sha256": "f327365aff670ef76700c3ef2a90093d73776bccad26166eb8ade6309a676dc6"}, "sigma_stat": 0.037, "sigma_syst": 0.033, "stat_basis": "measured — 1 sigma from the chi2 curvature of THIS band's fit (pipeline.fit_constraint.curvature_sigma, rescaled to red_chi2 = 1); NaN where the curvature was not measurable, never 0 and never the retired 1.000 clip"}

MOL-CN_AX_IR · N I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": null, "grade": "Codex Grade", "n_excluded": 0, "n_lines": 215, "provenance": {"artifact_mtime": "2026-09-13T16:56:48Z", "copied_to": "data/results/band_products/NI_MOL-CN_AX_IR_kpno_solar_atlas_solar_kpno_molecfits_corrected_SYNTH_MOL-CN_AX_IR_products.csv", "host": "Sirius", "ingested_at": "2026-09-13T16:57:06Z", "path": "/home/damienabraxas/rya1214_repo/data/results/band_products/NI_MOL-CN_AX_IR_kpno_solar_atlas_solar_kpno_molecfits_corrected_SYNTH_MOL-CN_AX_IR_products.csv", "sha256": "523ad48ba86edc1f03e8e1d23045bffa5844b14b2e48a9391781e46f393a7bf6"}, "sigma_stat": 0.046, "sigma_syst": 0.036, "stat_basis": "measured — 1 sigma from the chi2 curvature of THIS band's fit (pipeline.fit_constraint.curvature_sigma, rescaled to red_chi2 = 1); NaN where the curvature was not measurable, never 0 and never the retired 1.000 clip"}

Open provenance decisions

NIST's N I values ARE Tachiev (TP T7370). Adjudicating N I against this pull is one source seen twice, not independent confirmation.

[Adjudication record](#) · [Indicator gf checks](#)

Product visibility audit

Every feed entry is accounted for. Quarantined and archived entries retain their source disposition; their numeric fit outputs are available only in the labelled source download.

Withheld entries and source reasons (0)

Download the evidence

[Download PDF — Solar N appendix](#)

Source feed — includes quarantined fit outputs, not adopted abundances

Complete product visibility audit

Shared website/PDF report

Reproducibility manifest

Complete references and product-to-source bindings

Reproducibility

Generated from the pinned CNO campaign snapshot. Source commit

22ea65678043e016b12fb0defa596c10a200df88

Feed N.json v1.14 · updated 2026-09-25T19:39:46Z

Generated 2026-09-25T21:53:06-06:00 (source commit timestamp)

Feed SHA-256 9e4244db27dd4f2683e3cca67fd49973b9da6ac535edf171744ad5a23ec32fe2

Generator scripts/generate_cno_publication.py v1.0.0

[Pinned scientific source](#)

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar N atomic/molecular indicator lineage and atmospheric sensitivity; comparison literature, not an adopted Codex abundance. Sources: [1], [2].

Atomic data and gf authorities. The recorded N I/II pedigree is the partial NIST MCHF update described by Wiese & Fuhr; evaluated theory is not primary laboratory evidence. Sources: [3], [4].

Atomic data and gf authorities. NIST's N I values ARE Tachiev (TP T7370). Adjudicating N I against this pull is one source seen twice, not independent confirmation. Sources: [3], [5].

Model and grid sources. The N I registry uses the Amarsi GALAH/PySME 1D-NLTE departure grid. The solar 3D-NLTE atomic analysis is a separate source, not a full-3D correction applied by this feed. Sources: [6], [7], [8].

Atomic data and gf authorities. Atomic N I entry; source-family pedigree only, not a substitute for a per-transition gf join. Sources: [3], [4], [5].

Model and grid sources. Feed label: Synth · 1D-NLTE · Bergemann. Registered source: Amarsi/PySME N I 1D-NLTE. The generic Bergemann label is retained as supplied and is not used as a bibliographic attribution. Sources: [6], [7], [8].

Model and grid sources. Feed treatment: 1D-LTE; no NLTE/3D correction is claimed for this entry. Turbospectrum is the synthesis implementation. Sources: [9].

Molecular line data and delivery. CN line-data delivery: CN red A-X: Brooke et al. 2014 / Sneden et al. 2014 (isotopologues). Masseron compilation, VALD-based, distributed with iSpec/Turbospectrum (Gerber et al. 2023, A&A 669, A43; uploaded 2022 to keeper.mpd.l.mpg.de, split per wavelength region) — see molecules-dir README.md Sources: [2], [9], [10], [11].

Product-to-source index

N-products-92f4078d16e7 · products · red-optical · solar_jag · SET-AGSS21_ENGINE-A · ENGINE-A · not supplied · [1], [3], [4], [5], [6], [7], [8]

N-products-b6f3d1679545 · products · red-optical · solar_jag · SET-AGSS21 · 1D-LTE · not supplied · [1], [3], [4], [5], [9]

N-products-e3c293991e58 · products · red-optical · solar_kpno_kurucz2005_corrected · SET-AGSS21_ENGINE-A · ENGINE-A · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#), [\[7\]](#), [\[8\]](#)

N-products-08261af2d8eb · products · red-optical · solar_kpno_kurucz2005_corrected · SET-AGSS21 · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#)

N-products-e61a07d3c5f5 · products · red-optical · solar_kpno_molecfit_corrected · SET-AGSS21_ENGINE-A · ENGINE-A · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#), [\[7\]](#), [\[8\]](#)

N-products-dd0e12b67a86 · products · red-optical · solar_kpno_molecfit_corrected · SET-AGSS21 · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#)

N-products-a673b561de48 · products · red-optical · solar_iag · SET-AGSS21 · ENGINE-A · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#), [\[7\]](#), [\[8\]](#)

N-products-e57946a904b1 · products · red-optical · solar_kpno_kurucz2005_corrected · SET-AGSS21 · ENGINE-A · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#), [\[7\]](#), [\[8\]](#)

N-products-f95159b2157c · products · red-optical · solar_kpno_molecfit_corrected · SET-AGSS21 · ENGINE-A · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#), [\[7\]](#), [\[8\]](#)

N-products-150d2f215a8b · products · VIS · solar_harps_molecfit_corrected · MOL-CN_red · 1D-LTE · not supplied · [\[1\]](#), [\[2\]](#), [\[9\]](#), [\[10\]](#), [\[11\]](#)

N-products-9944e2b77baf · products · NIR · solar_iag · MOL-CN_AX_IR · 1D-LTE · not supplied · [\[1\]](#), [\[2\]](#), [\[9\]](#), [\[10\]](#), [\[11\]](#)

N-products-7fee3da2024e · products · NIR · solar_kpno_molecfit_corrected · MOL-CN_AX_IR · 1D-LTE · not supplied · [\[1\]](#), [\[2\]](#), [\[9\]](#), [\[10\]](#), [\[11\]](#)

Complete applicable bibliography

Solar abundance and indicator analyses

[1] — Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision. *A&A* 653, A141 DOI: 10.1051/0004-6361/202140445. [source record](#)

[2] — Amarsi et al. 2021, *A&A* 656 A113 DOI: 10.1051/0004-6361/202141384. [source record](#)

Atomic data and gf authorities

[3] — Kramida, A.; Ralchenko, Yu.; Reader, J.; NIST ASD Team (2024). NIST Atomic Spectra Database (version 5.11/5.12). National Institute of Standards and Technology, Gaithersburg, MD DOI: 10.18434/T4W30F. [source record](#)

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